

HQDFM Stack-up Impedance Report

Layer Count: 4

Stack-up Diagram:

L1	Finish 1oz(0.33oz+Plated)	Copper	1	0.0115mm	Residual Copp
	2116	pp	2116	0.1250mm	
L2/L3	1.2 H/H incl. copper	Core	incl. copper	1.2000mm	Residual Copp
	2116	pp	2116	0.1250mm	
L4	Finish 1oz(0.33oz+Plated)	Copper	1	0.0115mm	Residual Copp

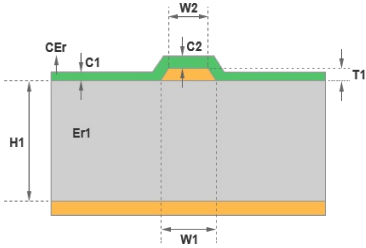
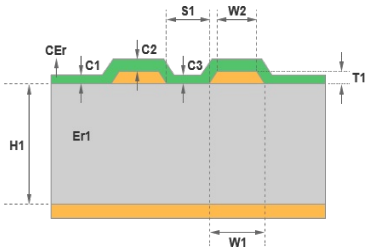
Board Thickness

Final:	1.6000mm +/-10%
Laminated:	1.4634 mm

Impedance Controlled Traces (Lengths in mil)

No.	Type	Layer	Orig. Trace Width	Orig. Trace Spacing	Orig. Distance To Copper	Upper Ref	Lower Ref	Trace Width	Trace Spacing	Distance To Copper	Target (ohm)	Tol.	Actual (ohm)
1	Outer Single-ended	L1	7.48	/	/	/	L2	7.90	/	/	50.00	(+/-10%)	49.65
2	Outer Differential	L1	8.66	5.95	/	/	L2	7.40	7.21	/	90.00	(+/-10%)	90.04

Impedance Calculations

Model	Values	Impedance Information
	W1 : 7.90 W2 : 7.10 H1 : 4.7323 Er1 : 4.2000 C1 : 1.1969 C2 : 0.5000 CEr : 3.5000	Outer Single-ended Controlled Layer: L1 Reference: &L Lower Trace Width: 7.90mil Required Impedance: 50.00 (+/-10%) Actual Impedance: 49.65 ohm
	W1 : 7.40 W2 : 6.60 S1 : 7.21 H1 : 4.7323 Er1 : 4.2000 T1 : 1.3780 C1 : 1.1969 C2 : 0.5000 C3 : 1.1969 CEr : 3.5000	Outer Differential Controlled Layer: L1 Reference: &L Lower Trace Width: 7.40mil Trace Separation: 7.21mil Required Impedance: 90.00 (+/-10%) Actual Impedance: 90.04 ohm